Question 1: The question is about Silver Iodide

The crystalline structure of $\beta$-AgI is similar to that of ice, allowing it to induce freezing by the process known as heterogeneous nucleation (cloud seeding). $\beta$-AgI is a bright yellow solid and has the wurtzite structure.

1) When the solid silver iodide is exposed to sunlight, it will darken rapidly. What is the oxidation state of silver in the darkened solid?

2) What is the solubility trend in AgF, AgCl, AgBr, AgI?

3) Consider:

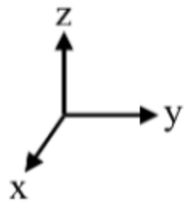
(a) $Ag^+ (aq) + e^- \longrightarrow Ag (s)$ $\qquad E^o = +0.80$ V

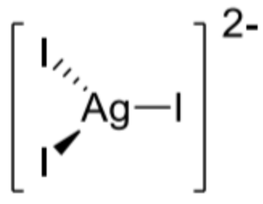
(b) $AgI (s) \rightleftharpoons Ag^+ (aq) + I^- (aq)$ $\qquad K_{sp} = 8.51 \times 10^{-17}$

(c) $Ag^+ (aq) + 3\ I^- (aq) \rightleftharpoons [AgI_3]^{2-} (aq)$ $\qquad K = 10^{14}$

From the given information, find the standard reduction potential of $[AgI_3]^{2-}$.

4) The salt $[PPh_3Me]_2[AgI_3]$ contains a triiodoargentate(I) ion $[AgI_3]^{2-}$ with approximate trigonal planar geometry. (See reference: Bowmaker, G. A.; Camus, A.; Skelton, B. W.; White, A. H. *J. Chem. Soc., Dalton Trans*., **1990**, 727-731.) The structure of $[AgI_3]^{2-}$ is shown below. Draw the crystal field splitting diagram for the *d* orbitals of silver and fill in all appropriate electrons.




Question 2: Compound Identification and Related Chemistry

A reaction between **M** and $Cl_2$ gives the only product $\mathbf{M}_x\mathbf{Cl}_y$. The following results are obtained under various conditions:

| **M (mol)** | **$Cl_2$ (mol)** | **Product (g)** |
|---|---|---|
| 0.20 | 0.80 | 26.7 |
| 0.30 | 0.70 | 40.0 |
| 0.40 | 0.60 | 53.3 |
| 0.50 | 0.50 | 44.4 |

| | | |
|---|---|---|
| 0.60 | 0.40 | 35.6 |
| 0.70 | 0.30 | 26.7 |
| 0.80 | 0.20 | 17.8 |

1) What is the chemical formula of $\mathbf{M}_x\mathrm{Cl}_y$? Identify **M**.

2) Write balanced chemical equations of:

(i) complete hydrolysis of $\mathbf{M}_x\mathrm{Cl}_y$

(ii) $\mathbf{M}_x\mathrm{Cl}_y + \mathrm{H_2SO_4}$

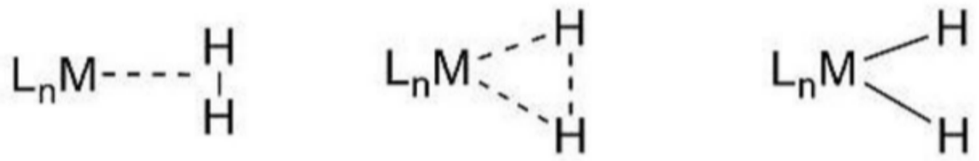
3) At certain temperatures, $\mathbf{M}_x\mathrm{Cl}_y$ exists in equilibrium as a dimer:

$$\mathbf{M}_x\mathrm{Cl}_y \rightleftharpoons (\mathbf{M}_x\mathrm{Cl}_y)_2$$

Draw a chemical structure of the dimer.

Question 3: Coordination chemistry of dihydrogen

Molecular hydrogen ($H_2$) can act as a ligand in certain transition metal complexes. In these complexes, the hydrogen atoms can have a different character depending on the metal as well as the nature of other ligands coordinated to the metal atom. They can range from a dihydrogen complex where the $H_2$ molecule keeps its bond, to dihydrido complexes when the bond between the two H atoms is completely broken. There are intermediate states as well, of course.




(a) Draw an MO energy diagram for the interaction between the metal d orbitals and the molecular orbitals of $H_2$. Indicate which orbitals on the metal are of correct symmetry to interact with each molecular orbital of $H_2$. Indicate how the energy levels of the orbitals change, and draw cartoon representations of these orbitals. Assume the $H_2$ bond/axis lies along the x-axis and the M-$H_2$ bond/axis lies along the z-axis.

Consider the two complexes below. One of them can be classified as a dihydrogen complex, whereas the other can be classified as a dihydrido complex.

$W(CO)_5H_2$ $Ir(PPh_3)_2H_2$

(b) Determine which complex belongs to which class. Provide a reason based on the MO diagrams you have drawn.

(c) Arrange the following complexes in order of decreasing H-H bond distance:

$W(CO)_5H_2$ $W(CO)_4(PPh_3)_2H$ $W(CO)_4(PPh_3)_2H_2$

(d) Draw the interactions of the ligand p-orbitals with the metal d-orbitals in M-X where X is a halide.

(e) Arrange the following complexes in order of decreasing H-H bond distance:

$Ir(CO)3(F)H_2$ $Ir(CO)_3(Cl)H_2$ $Ir(CO)_3(I)H_2$

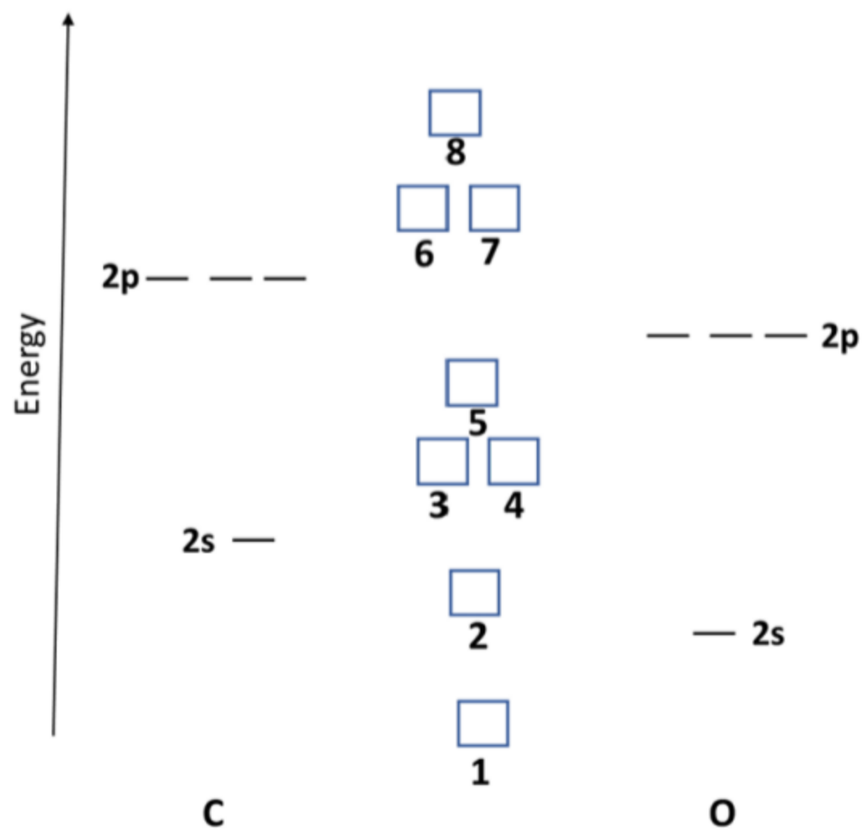
Question 4:

Ruthenium nanoparticles are efficient catalysts for CO combustion. The bonding interactions between CO molecules and the surface Ru atoms are of two types:

i. σ-bonding through a lone pair on the carbon atom;

ii. π-interaction through an overlap between the Ru valence d-orbitals and CO molecular orbitals.

1). Complete the MO diagram for CO with electrons and the following table with the type (σ/π) of overlap for forming each MO.



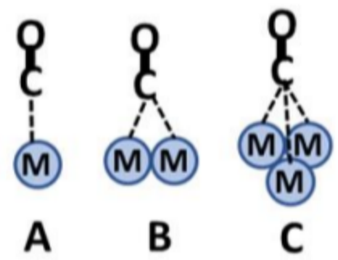
| MO | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Type of overlap | | | | | | | | |

2). What is the effect of the Ru− CO π-interaction on the Ru− CO and the C− O bond strengths? Tick the correct answer.

(i) Ru− CO bond strength: ☐ increases "in" ☐ decreases "dec" ☐ no change "nc"

(ii) C− O bond strength: ☐ increases "in" ☐ decreases "dec" ☐ no change "nc"

Adsorbed CO molecules have strong IR absorptions due to C− O stretching vibrations. Three binding modes for CO adsorbed on a metal surface are shown, where CO can interact with one or more metal atoms. IR peaks in the spectra of free CO and the three binding modes of adsorbed CO were observed at values of 1850, 1930, 2100, and 2143 $cm^{-1}$.




3). Tick which wavenumber corresponds to each adsorption mode (A, B, and C) and to free CO.

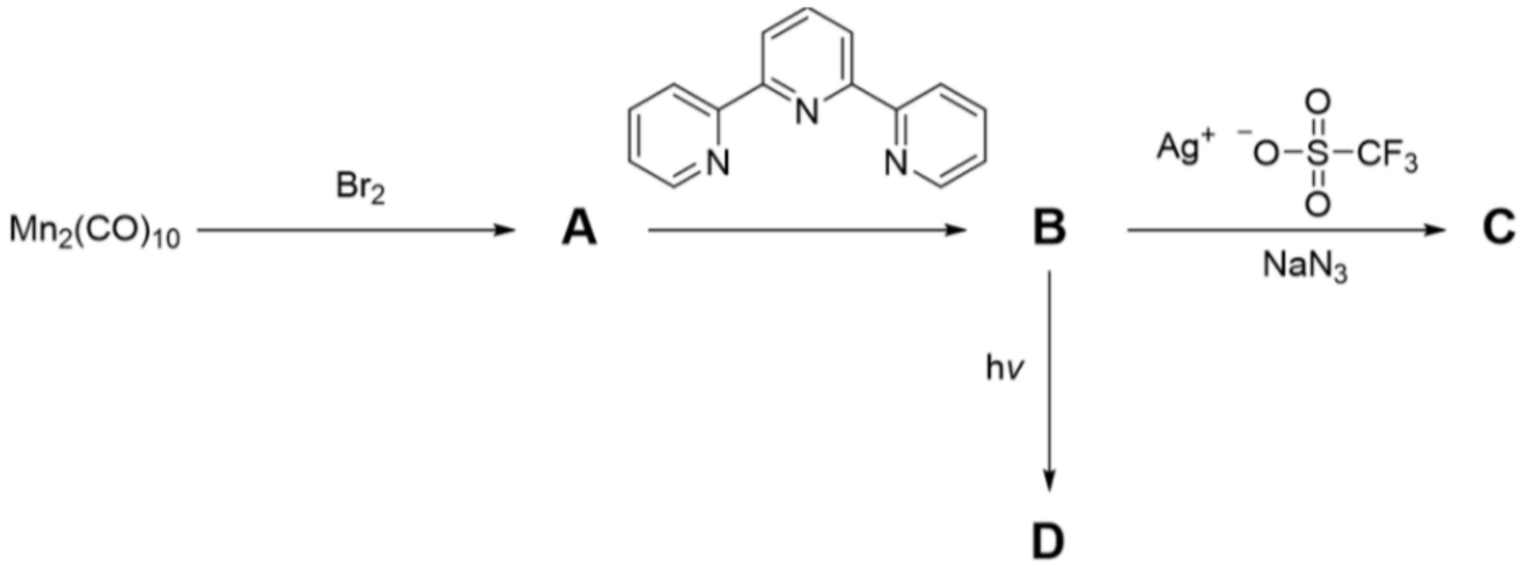
Despite being toxic at higher concentrations, at a concentration of 10-250 ppm, CO shows anti-inflammatory properties. Prodrugs known as CO releasing molecules (CORMs) target the delivery of effective dosages of CO to tissues in response to a trigger. Mn(I) carbonyl complexes have shown promise as PhotoCORMs as they can release CO when exposed to light.

Compounds A-D are all neutral Mn(I) carbonyl complexes.

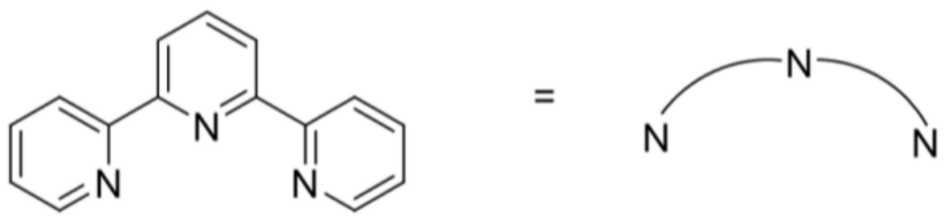
B, C, and D do not contain two CO ligands in a trans arrangement.

The IR spectrum of C contains a band at 2042 $cm^{-1}$ in addition to the CO stretching bands. The table with the number of CO stretching bands in the IR spectra of metal carbonyl complexes is given in the data sheet section at the beginning of the exam.

| Compound | %Mn by mass | Number of CO stretching bands |
|---|---|---|
| **A** | 19.98 | 3 |
| **B** | 12.15 | 3 |
| **C** | 13.26 | 3 |
| **D** | 12.95 | 2 |




For simplicity, the ligand can be drawn as:




4). Draw the structures of compounds A-D, ensuring the three-dimensional shape is clear. If the compound is chiral, only one enantiomer needs to be drawn.

Compound C reacts with alkynes and this can be used to tag proteins. This is known as an Inorganic Click (iClick) reaction.




During the conversion of C to E, the 2042 $cm^{-1}$ band in the IR spectrum of C disappears. The IR spectrum of E has two new C=O bands in the range of 1735-1725 $cm^{-1}$. After a further 6 h, compound E converts to compound F which only has one C=O band in the range of 1735-1725 $cm^{-1}$. Compound G does not contain manganese and has two planes of symmetry.

5). Draw the structures of compounds E, F, and G, ensuring the three-dimensional shape is clear. If the compound is chiral, only one enantiomer needs to be drawn.

3. Hydrogen is an environmentally clean fuel which is, in principle, ideal for powering the millions of vehicles on our roads.

1. **Write** the combustion reaction equations and **calculate** $\Delta_c H°$ at $T = 298$ K, $P = 1$ bar for:
   a) Hydrogen
   b) Methane
   c) Isooctane
2. **Determine** the energy density ($MJ \cdot kg^{-1}$) for these fuels at $T = 298$ K and $P = 1$ bar. **Choose** the substance that is the best fuel, considering the energy-to-mass ratio?

For the rest of this problem, assume that air consists of 20% $O_2$ and 80% $N_2$ by mole fraction.

3. **Determine** the volume of air at $T = 298$ K and $P = 1$ bar required for an internal combustion engine (ICE) operating on hydrogen fuel ($\eta = 36\%$) with a power of 1 kW to work for 1 hour.

Hydrogen-fuelled cars do not release $\mathbf{CO_2}$, in contrast to both gasoline and methane engines, which emit $CO_2$ as a direct product of fuel combustion. Uzbekistan maintains a national fleet of approximately **3.75 million cars**. **65% of them operate on methane, while 35% run on gasoline (assume that isooctane is used)**. An average vehicle in the country travels **22 000 km per year**.

*Note:*

- *Gasoline engines consume* ***9 dm³ of fuel per 100 km*** *($\rho_{gasoline} = 0.72\ kg \cdot dm^{-3}$).*
- *Methane engines consume* ***5 m³ of fuel per 100 km*** *($T = 298\ K$ and $P = 1\ bar$).*

4. **Estimate the amount of $CO_2$ emissions (in million tons) from cars that would be** avoided annually, if every vehicle in Uzbekistan operated solely on hydrogen.
5. **Calculate** the amount (in moles) of nitrogen contained in an isolated system with initial temperature, $T_{initial} = 298$ K, and constant pressure ($P = 1$ bar), containing 1 mol of $H_2$ and excess air, if after combustion, the temperature of the system reached $T = 800$ K.

When stoichiometric mixtures undergo combustion, the system temperature increases to such a high level that the combustion reaction begins to proceed in the reverse direction. As a result, a situation arises in which both the composition of the mixture and the final temperature after combustion are unknown.

6. In another isolated reactor, the process takes place under a constant pressure ($P = 1$ bar) and an initial temperature, $T_{initial} = 298$ K, with the assumption that only the combustion reaction takes place.
   a) **Determine** the final temperature for combustion of a stoichiometric mixture of hydrogen and oxygen, assuming that the reaction proceeds quantitatively.
   b) **Determine** the mole fractions of gases in the mixture using $K_p$ at the temperature from question 6-a

c) **Determine** the final temperature for the system using an energy balance, if after combustion the mixture composition corresponds to the mixture from question 6-b). **Compare** the final temperature with the temperature in 6-a).

d) **Determine** the real final temperature ($T_f$, with an accuracy of ~20 K) for the combustion of a stoichiometric $H_2$ and $O_2$ mixture. **Choose** the nearest temperature value for $T_f$:

i) 3900 K ii) 3850 K iii) 3800 K iv) 3750 K v) 3700 K vi) 3650 K

Combustion in air at high temperatures may lead to side reactions, such as NO formation.

**7.** **Calculate** the mole fraction of NO after the stoichiometric combustion of $H_2$ in air at equilibrium ($T = 3200$ K, $P_{total} = 1$ bar). Assume that only two reactions that reach equilibrium occur, and use reasonable assumptions where necessary.

**8.** **Choose** the option(s) which help to decrease NO formation in ICE operating on $H_2$ and air:

a) Lower the combustion temperature.

b) Increase the excess air ratio significantly.

c) Raise the compression ratio in the engine.

d) Increase the fuel flow speed.

For this problem, assume that $C_p$ does not depend on temperature, while enthalpy and entropy do depend.

Thermodynamic data ($T = 298$ K, $p = 1$ bar):

| **Substance** | **$\Delta_f H°$ / kJ·mol$^{-1}$** | **$S°$ / J·mol$^{-1}$·K$^{-1}$** | **$C_p$ / J·mol$^{-1}$·K$^{-1}$** |
|---|---|---|---|
| $H_2(g)$ | 0 | 130.7 | 28.8 |
| $O_2(g)$ | 0 | 205.2 | 29.4 |
| $N_2(g)$ | 0 | 191.6 | 29.1 |
| $H_2O(g)$ | –241.8 | 188.7 | 37.4 |
| $H_2O(l)$ | –285.8 | 70.0 | 75.4 |
| $CH_4(g)$ | –74.8 | 186.3 | 35.3 |
| $C_8H_{18}(l)$ | –260.2 | 361.2 | 254.7 |
| $CO_2(g)$ | –393.5 | 213.8 | 37.1 |
| $NO(g)$ | 90.3 | 210.8 | 29.9 |